Clark Zhang

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Work Authorization: Canadian Citizen — Eligible for U.S. TN Status (No H-1B required)

EDUCATION

University of British Columbia **September 2021 - April 2026**
Bachelor of Applied Science - Integrated Engineering (Mechatronics Specialization), Co-op Program
cGPA: 4.0 (92%)

- Relevant Courses: Mechanics III: 95%, Automatic Control Systems: 88%, Electromechanics: 94%, Thermodynamics: 100%, Heat Transfer: 94%, Fluid Mechanics: 91%

SKILLS

- Mechanical Design: SolidWorks (CSWP Certified, FEA, Flow), CATIA/3DX, Fusion 360, AutoCAD, Ansys FEA/Fluent, GD&T
- Programming & Electronics: Python, C++, PLC Control (UR/KUKA), PCB design (Altium), ROS2, Git, MATLAB/Simulink
- Prototyping & Manufacturing: FDM/SLA 3D printing, CNC machining, laser/waterjet cutting

WORK EXPERIENCE

Hein Labs **December 2024 – January 2026**
Engineering Project Lead - Mechatronics | SolidWorks, OpenCV, GD&T, ROS2, C++, Python, Altium

- Directed a 6-person engineering team developing a robotic drug-analysis platform for BC's overdose crisis response, automating sample preparation and downstream analysis of 1500+ samples with 98% overall reliability
- Designed, fabricated, and tested 3D-printable lab automation hardware in SolidWorks/Fusion 360, including a BLDC centrifuge, syringe-pump liquid handler, decapper, vial gripper, and robot transfer fixtures, reducing module costs from $1,500–$5,000 to <$100
- Produced 11 detailed CAD assemblies and manufacturing drawings with GD&T for precision, lab-grade components manufactured through 3D printing, CNC machining, and waterjet cutting
- Designed and fabricated 2 custom PCBs using Altium to achieve synchronized stepper and BLDC motor control
- Developed a multi-tiered control architecture using PLC/Python to orchestrate UR robotic arm motion and high-level workflows, integrated with C++/ROS2 for hardware module automation and sensor error detection

Hein Labs **April 2024 – December 2024**
Mechatronics Engineering Intern | SolidWorks, RoboDK, Python, C++, Git, PLC, I/O Wiring

- Commissioned 4 industrial robotic arms across 3 deployed self-driving chemistry lab platforms, handling robot mounting, I/O wiring, networking, and end-to-end system integration while reducing scientists' hands-on time by ~80%
- Transitioned hard-to-maintain PLC-based robot sequences to a reusable Python control library for UR and KUKA arms, making workflows easier to program, modify, and deploy
- Modeled 2 robotic platforms in SolidWorks and simulated workflows in RoboDK to verify reachability, collision clearance, and lab module access before deployment, reducing workflow errors by 5x

UBC AeroDesign **April 2022 - April 2026**
Airfoils Team Lead | CATIA, SolidWorks, ANSYS, AutoCAD, GD&T, Manufacturing, Composites

- Led an 11-person subteam designing and manufacturing a 120-inch heavy-lift RC aircraft for SAE AeroDesign competition, overseeing wing design, large-scale assembly integration, manufacturing, and testing. Earned 2nd place overall in 2025
- Designed main wing, ailerons, servo mounts, and linkage interfaces in SolidWorks and CATIA V5 using DFMA and GD&T principles, allowing easy integration into overall assembly
- Optimized aircraft geometry through SolidWorks FEA/Flow, ANSYS, and wind-tunnel validation, improving lift performance by ~20% and verifying structural stiffness before manufacturing
- Manufactured flight hardware using 3D printing, CNC machining, waterjet/laser cutting, and carbon-fiber prepreg layups, standardizing assembly processes and reducing build errors by 70%

PROJECTS

Robotic Tool Vending Machine (Haven) **September 2025 - April 2026**
Project Lead | SolidWorks, Raspberry Pi, Klipper, KiCAD, Manufacturing & Prototyping

- Designed and validated a Cartesian XY gantry and rotary storage system in SolidWorks FEA before fabrication.
- Fabricated and integrated prototypes using 3D printing, waterjet cutting, hand tools, and CNC machining
- Developed Raspberry Pi/Klipper motion control with magnetic encoder feedback and custom I2C multiplexer PCB designed in KiCAD to synchronize gantry travel and storage indexing

Semi-Autonomous Aquatic Trash-Removal Robot **September 2024 - April 2025**
Mechanical Lead | SolidWorks, 3D-printing, Waterjet, Machine Shop Tools

- Designed aquatic mechanical assemblies in SolidWorks, engineering custom dual-pontoon flotation chassis, structural supports, and adjustable propulsion mounts. Won 1st place at the National CSME Design Competition
- Fabricated and integrated sealed plywood, aluminum, and 3D-printed components to assemble a watertight product optimized for saltwater operations